

41. Let $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$. If $\int_0^1 (1-x^{10})^{20} dx = a \times \beta(b, c)$, then $100(a+b+c)$ equals :

माना $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$ है। यदि $\int_0^1 (1-x^{10})^{20} dx = a \times \beta(b, c)$ है, तो $100(a+b+c)$ बराबर है :

(A) 1021

(B) 2120

(C) 2012

(D) 1120

Ans. B

Sol. $I = \int_0^1 (1-x^{10})^{20} dx$

$$x^{10} = t \Rightarrow x = (t)^{1/10}$$

$$I = \frac{1}{10} \int_0^1 (t)^{-9/10} (1-t)^{20} dt$$

$$a = \frac{1}{10}, b = \frac{1}{10}, c = 21$$

$$100 \left(\frac{1}{10} + \frac{1}{10} + 21 \right)$$

$$\Rightarrow 10 + 10 + 2100 = 2120$$

42. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx$ is equal to :

$$\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx \text{ बराबर है :}$$

(A) 1/9

(B) 1/6

(C) 1/3

(D) 1/12

Ans. B

Sol. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx$

$$\Rightarrow \int_0^{\pi/4} \frac{\tan^2 x \cdot \sec^2 x}{(1 + \tan^3 x)^2} \cdot dx$$

$$\text{Let } 1 + \tan^3 x = t$$

$$3 \cdot \tan^2 x \cdot \sec^2 x \cdot dx = dt$$

$$\Rightarrow \frac{1}{3} \int_1^2 \frac{dt}{(t^2)^2}$$

$$\Rightarrow \frac{1}{3} \left[-\frac{1}{t} \right]_1^2$$

$$\Rightarrow -\frac{1}{3} \left[+\frac{1}{2} - 1 \right] \Rightarrow \frac{1}{6}$$

43. The value of $k \in \mathbb{N}$ for which the integral $I_n = \int_0^1 (1-x^k)^n dx$, $n \in \mathbb{N}$, satisfies $147 I_{20} = 148 I_{21}$ is :

$k \in \mathbb{N}$ का मान जिसके लिए समाकलन $I_n = \int_0^1 (1-x^k)^n dx$, $n \in \mathbb{N}$ एवं $147 I_{20} = 148 I_{21}$ को संतुष्ट करता है :

(A) 8

(B) 7

(C) 10

(D) 14

Ans. B

Sol. $I_n = \int_0^1 (1-x^k)^n dx$

$$I_{21} = \int_0^1 (1-x^k)^{21} dx$$

$$\Rightarrow \int_0^1 (1-x^k) (1-x^k)^{20} dx$$

$$\Rightarrow \int_0^1 (1-x^k)^{20} dx - \int_0^1 x^k (1-x^k)^{20} dx$$

$$I_{21} = I_{20} - \int_0^1 x^k (1-x^k)^{20} dx$$

$$\int_0^1 x x^{(k-1)} (1-x^k)^{20} dx$$

$$1-x^k = t$$

$$-K(x^{K-1} dx) = dt$$

Integration by part

$$x \left[-\frac{(1-x^K)^{21}}{21K} \right]_0^1 - \frac{1}{21K} \int_0^1 (1-x^k)^{21} dx$$

$$I_{21} = I_{20} + 0 - \frac{I_{21}}{21K}$$

$$= \left(\frac{21K+1}{21K} \right) I_{21} = I_{20}$$

$$= (21K+1) I_{21} = 21KI_{20}$$

$$K = 7$$

44. Let $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$. Then e^α and $e^{-\alpha}$ are the roots of the equation :

माना $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$ है। तो e^α तथा $e^{-\alpha}$ निम्न में से किस समीकरण के मूल हैं :

(A) $2x^2 - 5x - 2 = 0$ (B) $x^2 - 2x - 8 = 0$ (C) $2x^2 - 5x + 2 = 0$ (D) $x^2 + 2x - 8 = 0$

Ans. C

Sol. $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$

Let $e^x - 1 = t^2$

$e^x dx = 2t dt$

$$\int \frac{2t dt}{e^x \cdot t} = \int \frac{2 dt}{t^2 + 1}$$

$$= 2 \tan^{-1} t$$

$$= 2 \tan^{-1} \left(\sqrt{e^x - 1} \right)_{\alpha}^{\log_e 4}$$

$$= 2 \left[\tan^{-1} \sqrt{3} - \tan^{-1} \sqrt{e^\alpha - 1} \right] = \frac{\pi}{6}$$

$$= \frac{\pi}{3} - \tan^{-1} \sqrt{e^\alpha - 1} = \frac{\pi}{12}$$

$$= \tan^{-1} \sqrt{e^\alpha - 1} = \frac{\pi}{4}$$

$$e^\alpha = 2, \quad e^{-\alpha} = \frac{1}{2}$$

$$x^2 - \left(2 + \frac{1}{2}\right)x + 1 = 0$$

$$2x^2 - 5x + 2 = 0$$

45. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin \left(2t^{\frac{1}{3}} \right) + \cos \left(t^{\frac{1}{3}} \right) \right) dt}{\left(x - \frac{\pi}{2} \right)^2} \right)$ is equal to :

$\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin \left(2t^{\frac{1}{3}} \right) + \cos \left(t^{\frac{1}{3}} \right) \right) dt}{\left(x - \frac{\pi}{2} \right)^2} \right)$ बराबर है :

(A) $\frac{11\pi^2}{10}$

(B) $\frac{3\pi^2}{2}$

(C) $\frac{5\pi^2}{9}$

(D) $\frac{9\pi^2}{8}$

Ans. D

Sol. L. hospital's rule

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} \frac{-3x^2 \cdot (\sin 2x + \cos x)}{2 \left(x - \frac{\pi}{2} \right)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-3x^2 \cos x (2 \sin x + 1)}{2 \left(x - \frac{\pi}{2} \right)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-9 \times \frac{\pi^2}{4}}{2} \times \frac{\cos x}{x - \frac{\pi}{2}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-9\pi^2}{8} \times \frac{-\sin x}{1}$$

$$= \frac{9\pi^2}{8}$$

46. The integral $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos \left(2 \cot^{-1} \sqrt{\frac{1-x}{1+x}} \right) dx$ is equal to :

समाकलन $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos \left(2 \cot^{-1} \sqrt{\frac{1-x}{1+x}} \right) dx$ बराबर है :

(A) $\frac{1}{4}$

(B) $-\frac{1}{2}$

(C) $-\frac{1}{4}$

(D) $\frac{1}{2}$

Ans. C

Sol. $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos \left(2 \cot^{-1} \sqrt{\frac{1-x}{1+x}} \right) dx$

$$x = \cos 2\theta$$

$$\Rightarrow dx = (-2 \sin 2\theta d\theta)$$

Take limit as α and β

$$-2 \int_{\alpha}^{\beta} \cos 2\theta \cdot \sin 2\theta d\theta$$

$$= \int_{\alpha}^{\beta} \sin 4\theta d\theta$$

$$= \frac{-\cos 4\theta}{4} \Big|_{\alpha}^{\beta}$$

$$= -\frac{1}{4} \left(2 \cdot (x^2) - 1 \right) \Big|_{1/4}^{3/4}$$

$$= -\frac{1}{4} \left(2x^2 - 1 \right) \Big|_{1/4}^{3/4}$$

$$= -\frac{1}{4} \left(\frac{18}{16} - 1 - \frac{2}{16} + 1 \right)$$

$$= -\frac{1}{4}$$

47. The value of the integral $\int_{-1}^2 \log_e (x + \sqrt{x^2 + 1}) dx$ is :

समाकलन $\int_{-1}^2 \log_e (x + \sqrt{x^2 + 1}) dx$ का मान है :

- (A) $\sqrt{2} - \sqrt{5} + \log_e \left(\frac{7 + 4\sqrt{5}}{1 + \sqrt{2}} \right)$ (B) $\sqrt{5} - \sqrt{2} + \log_e \left(\frac{9 + 4\sqrt{5}}{1 + \sqrt{2}} \right)$
- (C) $\sqrt{2} - \sqrt{5} + \log_e \left(\frac{9 + 4\sqrt{5}}{1 + \sqrt{2}} \right)$ (D) $\sqrt{5} - \sqrt{2} + \log_e \left(\frac{7 + 4\sqrt{5}}{1 + \sqrt{2}} \right)$

Ans. C

Sol. $\int_{-1}^2 \log_e (x + \sqrt{x^2 + 1}) dx$

$$= \left[x \log_e (x + \sqrt{x^2 + 1}) \right]_{-1}^2 - \int_{-1}^2 \frac{x}{(x + \sqrt{x^2 + 1})} \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right) dx$$

$$= 2 \log_e (2 + \sqrt{5}) + \log_e (\sqrt{2} - 1) - \int_{-1}^2 \frac{x}{\sqrt{x^2 + 1}} dx$$

$$= \log_e \left[(2 + \sqrt{5})^2 (\sqrt{2} - 1) \right] - \left[\sqrt{x^2 + 1} \right]_{-1}^2$$

$$= \log_e \left[\frac{9 + 4\sqrt{5}}{1 + \sqrt{2}} \right] - \sqrt{5} + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{5} + \log_e \left[\frac{9 + 4\sqrt{5}}{1 + \sqrt{2}} \right]$$

48. If (a, b) be the orthocentre of the triangle whose vertices are (1,2), (2,3) and (3,1), and $I_1 = \int_a^b x \sin (4x - x^2) dx$,

$I_2 = \int_a^b \sin(4x - x^2) dx$, then $36 \frac{I_1}{I_2}$ is equal to :

यदि शीर्षों (1,2), (2,3) तथा (3,1) के त्रिभुज का लंबकेन्द्र (a, b) है, तथा $I_1 = \int_a^b x \sin(4x - x^2) dx$, $I_2 = \int_a^b \sin(4x - x^2) dx$ है,

तो $36 \frac{I_1}{I_2}$ बराबर है :

(A) 72

(B) 88

(C) 66

(D) 80

Ans. A

Sol. Slope of AB = 1.

Altitude through C

$$y - 1 = -1(x - 3)$$

$$\Rightarrow x + y = 4.$$

Orthocentre (a, b) lies on this

$$\Rightarrow a + b = 4 \quad \text{_____ (i)}$$

$$\text{Now, } I_1 = \int_a^b x \cdot \sin(4x - x^2) dx$$

$$= \int_a^b (a + b - x) \sin(4(a + b - x) - (a + b - x)^2) dx$$

$$= \int_a^b (4 - x) \sin(4x - x^2) dx \quad \{\text{using (i)}\}$$

$$\Rightarrow I_1 = 4 \cdot I_2 - I_1$$

$$\Rightarrow \frac{I_1}{I_2} = 2$$

$$\Rightarrow 36 \frac{I_1}{I_2} = 72$$

49. For $0 < a < 1$, the value of the integral $\int_0^\pi \frac{dx}{1 - 2a \cos x + a^2}$ is :

$0 < a < 1$ के लिए, समाकलन $\int_0^\pi \frac{dx}{1 - 2a \cos x + a^2}$ का मान है :

(A) $\frac{\pi}{1-a^2}$

(B) $\frac{\pi^2}{\pi+a^2}$

(C) $\frac{\pi^2}{\pi-a^2}$

(D) $\frac{\pi}{1+a^2}$

Ans. A

Sol. $I = \int_0^\pi \frac{dx}{1-2a \cos x + a^2}; 0 < a < 1$

$$I = \int_0^\pi \frac{dx}{1+2a \cos x + a^2}$$

$$2I = 2 \int_0^{\pi/2} \frac{2(1+a^2)}{(1+a^2)^2 - 4a^2 \cos^2 x} dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2(1+a^2) \cdot \sec^2 x}{(1+a^2)^2 \cdot \sec^2 x - 4a^2} dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2 \cdot (1+a^2) \cdot \sec^2 x}{(1+a^2)^2 \cdot \tan^2 x + (1-a^2)^2} dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\frac{2 \cdot \sec^2 x}{1+a^2} \cdot dx}{\tan^2 x + \left(\frac{1-a^2}{1+a^2}\right)^2}$$

$$\Rightarrow I = \frac{2}{(1-a^2)} \left[\frac{\pi}{2} - 0 \right]$$

$$I = \frac{\pi}{1-a^2}$$

50. If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{x^2 \cos x}{1+\pi^x} + \frac{1+\sin^2 x}{1+e^{\sin x^{2023}}} \right) dx = \frac{\pi}{4}(\pi+a) - 2$, then the value of a is :

यदि समाकलन $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{x^2 \cos x}{1+\pi^x} + \frac{1+\sin^2 x}{1+e^{\sin x^{2023}}} \right) dx = \frac{\pi}{4}(\pi+a) - 2$ है, तो a का मान है :

(A) $\frac{3}{2}$

(B) 3

(C) 2

(D) $-\frac{3}{2}$

Ans. B

Sol. By even odd property

$$\int_0^{\frac{\pi}{2}} (x^2 \cos x + \sin^2 x + 1) dx$$

$$\int_0^{\frac{\pi}{2}} \left(x^2 \cos x + \frac{3}{2} - \frac{\cos 2x}{2} \right) dx$$

$$\left(x^2 \sin x - \int 2x \sin x dx + \frac{3x}{2} - \frac{\sin 2x}{4} \right)_0^{\frac{\pi}{2}}$$

$$= \left[x^2 \sin x - \left[2x(-\cos x) - \int 2(-\cos x) dx \right] + \frac{3x}{2} - \frac{\sin 2x}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \left[x^2 \sin x + 2x \cos x - 2 \sin x + \frac{3x}{2} - \frac{\sin 2x}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \left[\frac{\pi^2}{4} + 0 - 2 + \frac{3\pi}{4} - 0 \right] - [0] = \frac{\pi^2}{4} + \frac{3\pi}{4} - 2$$

51. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1}{\left(x - \frac{\pi}{2}\right)^2} \int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos \left(t^{\frac{1}{3}}\right) dt \right)$ is equal to :

$\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1}{\left(x - \frac{\pi}{2}\right)^2} \int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos \left(t^{\frac{1}{3}}\right) dt \right)$ का मान है :

(A) $\frac{3\pi}{4}$

(B) $\frac{3\pi^2}{8}$

(C) $\frac{3\pi^2}{4}$

(D) $\frac{3\pi}{8}$

Ans. B

Sol. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\frac{\pi}{2}}^x \cos(t^3) dt}{\left(x - \frac{\pi}{2}\right)^2} = \frac{0}{0}$ form

Apply leibniz

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos\left(\frac{\pi}{2}\right) \times 0 - \cos(x) \cdot 3x^2}{2\left(x - \frac{\pi}{2}\right)}$$

$$x \rightarrow \frac{\pi}{2} + t$$

$$\Rightarrow t \rightarrow 0$$

$$\lim_{t \rightarrow 0} \frac{-\cos\left(\frac{\pi}{2} + t\right) 3\left(\frac{\pi}{2}\right)^2}{2t}$$

$$\lim_{t \rightarrow 0} \frac{3\pi^2 \sin t}{8t} = \frac{3\pi^2}{8}$$

52. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n^3}{(n^2 + k^2)(n^2 + 3k^2)}$ is :

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n^3}{(n^2 + k^2)(n^2 + 3k^2)} \text{ का मान है :}$$

(A) $\frac{(2\sqrt{3} + 3)\pi}{24}$ (B) $\frac{\pi}{8(2\sqrt{3} + 3)}$ (C) $\frac{13(2\sqrt{3} - 3)\pi}{8}$ (D) $\frac{13\pi}{8(4\sqrt{3} + 3)}$

Ans. D

Sol. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{n^3}{n^4 \left(1 + \frac{k^2}{n^2}\right) \left(1 + \frac{3k^2}{n^2}\right)}$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{n^3}{\left(1 + \frac{k^2}{n^2}\right) \left(1 + \frac{3k^2}{n^2}\right)}$$

$$= \int_0^1 \frac{dx}{3(1+x^2) \left(\frac{1}{3} + x^2\right)}$$

$$= \frac{1}{3} \times \frac{3}{2} \int_0^1 \frac{(x^2 - 1) - \left(x^2 + \frac{1}{3}\right)}{(1+x^2) \left(\frac{1}{3} + x^2\right)} dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{1}{x^2 + \left(\frac{1}{\sqrt{3}}\right)^2} - \frac{1}{1+x^2} \right] dx$$

$$= \frac{1}{2} \left[\sqrt{3} \tan^{-1}(\sqrt{3}x) \right]_0^1 - \frac{1}{2} \left(\tan^{-1} x \right)_0^1$$

$$= \frac{\sqrt{3}}{2} \left(\frac{\pi}{3} \right) - \frac{1}{2} \left(\frac{\pi}{4} \right)$$

$$= \frac{\pi}{2\sqrt{3}} - \frac{\pi}{8}$$

$$= \frac{\pi}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{4} \right)$$

$$= \frac{\pi}{2} \left(\frac{4 - \sqrt{3}}{4\sqrt{3}} \right) \left(\frac{4 + \sqrt{3}}{4 + \sqrt{3}} \right)$$

$$= \frac{\pi}{2} \left(\frac{16 - 3}{4\sqrt{3}(4 + \sqrt{3})} \right)$$

$$= \frac{13\pi}{8(4\sqrt{3} + 3)}$$

53. Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = \frac{1}{2}$. If the $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{e^{x^2} - 1} = \alpha$, then $8\alpha^2$ is equal to :

माना $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ एक अवकलनीय फलन है तथा $f(0) = \frac{1}{2}$ है। यदि $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{e^{x^2} - 1} = \alpha$ है, तो $8\alpha^2$ बराबर है :

- (A) 16 (B) 2 (C) 4 (D) 1

Ans. B

Sol.
$$\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt + xf(x)}{e^{x^2} (2x)} = \lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{e^{x^2} (2x)} + \lim_{x \rightarrow 0} \frac{f(x)}{e^{x^2} (2)}$$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{e^{x^2} + xe^{x^2} (2x)} + \frac{1}{4} = \frac{1}{2}$$

$$\Rightarrow 8\alpha^2 = 2$$

54. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = ae^{2x} + be^x + cx$. If $f(0) = -1$, $f'(\log_e 2) = 21$ and $\int_0^{\log_e 4} (f(x) - cx) dx = \frac{39}{2}$, then the value of $|a + b + c|$ equals :

माना Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = ae^{2x} + be^x + cx$ द्वारा परिभाषित है। यदि $f(0) = -1$, $f'(\log_e 2) = 21$ तथा

$\int_0^{\log_e 4} (f(x) - cx) dx = \frac{39}{2}$ हैं, तो $|a + b + c|$ का मान बराबर है :

- (A) 12 (B) 16 (C) 8 (D) 10

Ans. C

Sol. $f(x) = ae^{2x} + be^x + cx$ $f(0) = -1$
 $a + b = -1$
 $f'(x) = 2ae^{2x} + be^x + c$ $f(\ln 2) = 21$
 $8a + 2b + c = 21$

$$\int_0^{\ln 4} (ae^{2x} + be^x) dx = \frac{39}{2}$$

$$\left[\frac{ae^{2x}}{2} + be^x \right]_0^{\ln 4} = \frac{39}{2} \Rightarrow 8a + 4b - \frac{a}{2} - b = \frac{39}{2}$$

$$15a + 6b = 39$$

$$15a - 6a - 6 = 39$$

$$9a = 45 \Rightarrow a = 5$$

$$b = -6$$

$$c = 21 - 40 + 12 = -7$$

$$a + b + c = -8$$

$$|a + b + c| = 8$$

55. Let a and b be real constants such that the function f defined by $f(x) = \begin{cases} x^2 + 3x + a & , x \leq 1 \\ bx + 2 & , x > 1 \end{cases}$ be differentiable on

$\mathbb{R}$. Then, the value of $\int_{-2}^2 f(x)dx$ equal :

माना वास्तविक अचर a तथा b इस प्रकार हैं कि फलन $f(x) = \begin{cases} x^2 + 3x + a & , x \leq 1 \\ bx + 2 & , x > 1 \end{cases}$, $\mathbb{R}$ पर अवकलनीय है। तो

$\int_{-2}^2 f(x)dx$ का मान बराबर है :

(A) 21

(B) $\frac{15}{6}$

(C) 17

(D) $\frac{19}{6}$

Ans. C

Sol. $f(x)$ is continuous at $x = 1$

$$f(1^-) = f(1) = f(1^+)$$

$$a + 4 = b + 2 \Rightarrow b = a + 2$$

$f(x)$ is differential at $x = 1$

$$f'(1^-) = f'(1^+)$$

$$5 = b \Rightarrow a = 3$$

$$\text{Now } \int_{-2}^2 f(x)dx = \int_{-2}^1 (x^2 + 3x + 3)dx + \int_1^2 (5x + 2)dx$$

$$\begin{aligned}
 &= \left(\frac{x^3}{3} + \frac{3x^2}{2} + 3x \right)_{-2}^1 + \left(\frac{5x^2}{2} + 2x \right)_1^2 = \left(\frac{1}{3}(1+8) + \frac{3}{2}(1-4) + 9 \right) + \frac{5}{2}(4-1) + 2 \\
 &= \frac{1}{3} + \frac{3}{2} + 3 - \left(\frac{-8}{3} + 6 - 6 \right) + \left(10 + 4 - \frac{5}{2} - 2 \right) \\
 &= 6 + \frac{3}{2} + 12 - \frac{5}{2} = 18 - 1 = 17
 \end{aligned}$$

56. Let $y = f(x)$ be a thrice differentiable function in $(-5, 5)$. Let the tangents to the curve $y = f(x)$ at $(1, f(1))$ and $(3, f(3))$ make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive x-axis. If $27 \int_1^3 \left((f'(t))^2 + 1 \right) f''(t) dt = \alpha + \beta\sqrt{3}$ where α, β are integers, then the value of $\alpha + \beta$ equals :

माना फलन $y = f(x)$ अंतराल $(-5, 5)$ तीन बार अवकलनीय है। माना वक्र $y = f(x)$ के बिन्दुओं $(1, f(1))$ तथा $(3, f(3))$ पर स्पर्श रेखाएँ धनात्मक x-अक्ष से $\frac{\pi}{6}$ तथा $\frac{\pi}{4}$ के कोण बनाती हैं। यदि $27 \int_1^3 \left((f'(t))^2 + 1 \right) f''(t) dt = \alpha + \beta\sqrt{3}$ हैं, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ का मान बराबर है :

- (A) 26 (B) -16 (C) -14 (D) 36

Ans. A

Sol. $y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$

$$\left. \frac{dy}{dx} \right|_{(1, f(1))} = f'(1) = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \Rightarrow f'(1) = \frac{1}{\sqrt{3}}$$

$$\left. \frac{dy}{dx} \right|_{(3, f(3))} = f'(3) = \tan \frac{\pi}{4} = 1 \Rightarrow f'(3) = 1$$

$$27 \int_1^3 \left((f'(t))^2 + 1 \right) f''(t) dt = \alpha + \beta\sqrt{3}$$

$$I = \int_1^3 \left((f'(t))^2 + 1 \right) f''(t) dt$$

$$f'(t) = z \Rightarrow f''(t) dt = dz$$

$$z = f'(3) = 1$$

$$z = f'(1) = \frac{1}{\sqrt{3}}$$

$$I = \int_{\frac{1}{\sqrt{3}}}^1 (z^2 + 1) dz = \left(\frac{z^3}{3} + z \right)_{\frac{1}{\sqrt{3}}}^1$$

$$= \left(\frac{1}{3} + 1 \right) - \left(\frac{1}{3 \cdot 3\sqrt{3}} + \frac{1}{\sqrt{3}} \right)$$

$$= \frac{4}{3} - \frac{10}{9\sqrt{3}} = \frac{4}{3} - \frac{10}{27}\sqrt{3}$$

$$\alpha + \beta\sqrt{3} = 27 \left(\frac{4}{3} - \frac{10}{27}\sqrt{3} \right) = 36 - 10\sqrt{3}$$

$$\alpha = 36, \beta = -10$$

$$\alpha + \beta = 36 - 10 = 26$$

57. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x}{(1+x^4)^{1/4}}$ and $g(x) = f(f(f(f(x))))$. Then, $18 \int_0^{\sqrt{2\sqrt{5}}} x^2 g(x) dx$ equal to :

माना फलन $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{x}{(1+x^4)^{1/4}}$ द्वारा परिभाषित है तथा $g(x) = f(f(f(f(x))))$ है। तो $18 \int_0^{\sqrt{2\sqrt{5}}} x^2 g(x) dx$

बराबर है :

(A) 42

(B) 33

(C) 36

(D) 39

Ans. D

Sol.
$$f(x) = \frac{x}{(1+x^4)^{1/4}}$$

$$fof(x) = \frac{f(x)}{(1+f(x)^4)^{1/4}} = \frac{\frac{x}{(1+x^4)^{1/4}}}{\left(1 + \frac{x^4}{1+x^4}\right)^{1/4}} = \frac{x}{(1+2x^4)^{1/4}}$$

$$f(f(f(f(x)))) = \frac{x}{(1+4x^4)^{\frac{1}{4}}}$$

$$18 \int_0^{\sqrt[4]{2\sqrt{5}}} \frac{x^3}{(1+4x^4)^{\frac{1}{4}}} dx$$

$$\text{Let } 1 + 4x^4 = t^4$$

$$16x^3 dx = 4t^3 dt$$

$$\frac{18}{4} \int_1^3 \frac{t^3 dt}{t}$$

$$= \frac{9}{2} \left(\frac{t^3}{3} \right)_1^3$$

$$= \frac{3}{2} [26] = 39$$

58. Let a be the sum of all coefficients in the expansion of $(1 - 2x + 2x^2)^{2023} (3 - 4x^2 + 2x^3)^{2024}$ and

$$b = \lim_{x \rightarrow 0} \left(\frac{\int_0^x \frac{\log(1+t)}{t^{2024} + 1} dt}{x^2} \right). \text{ If the equations } cx^2 + dx + e = 0 \text{ and } 2bx^2 + ax + 4 = 0 \text{ have a common root, where c,}$$

d, e ∈ R, then d : c : e equals :

$$\text{माना } (1 - 2x + 2x^2)^{2023} (3 - 4x^2 + 2x^3)^{2024} \text{ के प्रसार में सभी पदों के गुणांक का योग } a \text{ है तथा } b = \lim_{x \rightarrow 0} \left(\frac{\int_0^x \frac{\log(1+t)}{t^{2024} + 1} dt}{x^2} \right) \text{ है।}$$

यदि समीकरणों $cx^2 + dx + e = 0$ तथा $2bx^2 + ax + 4 = 0$ का एक उभयनिष्ठ मूल है, जहाँ c, d, e ∈ R हैं, तो d : c : e बराबर है :

- (A) 1 : 1 : 4 (B) 4 : 1 : 4 (C) 2 : 1 : 4 (D) 1 : 2 : 4

Ans. A

Sol. By putting x = 1

$$a = (1 - 2 + 2)^{2023} (3 - 4 + 2)^{2024} = (1) (1) = 1$$

$$b = \lim_{x \rightarrow 0} \frac{\int_0^x \frac{\ln(1+t)}{t^{2024}+1} dt}{x^2} \left(\frac{0}{0} \text{ form} \right)$$

By DL hospital Rule

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\frac{\ln(1+x)}{x^{2024}+1}}{2x} \\ &= \frac{1}{2} \cdot \frac{1}{(0+1)} = \frac{1}{2} \end{aligned}$$

Now $cx^2 + dx + e = 0$ and $x^2 + x + 4 = 0$ have a common root.

But $D < 0$ for second equation. So both roots are common.

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{1}$$

$$d : c : e = 1 : 1 : 4$$

59. Let $f, g : (0, \infty) \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt$ and $g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$. Then the value of $9 \left(f(\sqrt{\log_e 9}) + g(\sqrt{\log_e 9}) \right)$ is equal to :

माना $f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt$ तथा $g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$ द्वारा परिभाषित दो फलन $f, g : (0, \infty) \rightarrow \mathbb{R}$ हैं। तो $9 \left(f(\sqrt{\log_e 9}) + g(\sqrt{\log_e 9}) \right)$ का मान बराबर है :

- (A) 9 (B) 6 (C) 10 (D) 8

Ans. D

Sol. $f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt$
 $\Rightarrow f'(x) = 2 \cdot (|x| - x^2) e^{-x^2} \dots\dots\dots(1)$

$$g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$$

$$g'(x) = x^{-x^2} (2x) - 0$$

$$f'(x) + g'(x) = 2x^{-x^2} - 2x^2 e^{-x^2} + 2x^2 e^{-x^2}$$

Integrating both sides w.r.t.x

$$f(x) + g(x) = \int_0^{\alpha} 2xe^{-x^2} dx$$

$$x^2 = t$$

$$\Rightarrow \int_0^{\sqrt{\alpha}} e^{-t} dt = \left[-e^{-t} \right]_0^{\sqrt{\alpha}}$$

$$= -e^{(\log_e(9))^{-1}+1}$$

$$\Rightarrow 9(f(x) + g(x)) = \left(1 - \frac{1}{9}\right)9 = 8$$

60. The value of $\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} dx$ is equal to :

$$\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} dx \text{ का मान बराबर है :}$$

(A) 1

(B) 2

(C) 0

(D) -1

Ans. C

Sol. $2x^3 - 3x^2 - x + 1 = \frac{1}{4}(2x-1)((2x-1)^2 - 5)$

$$I = \int_0^1 \left(\frac{(2x-1)((2x-1)^2 - 5)}{4} \right)^{\frac{1}{3}} dx$$

put $2x - 1 = t$

$$I = \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{t(t^2 - 5)}{4} \right)^{\frac{1}{3}} dt$$

$$= 0 (\because \text{odd function in } t)$$